

Jiaming Lu

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Student Pass holder | Available Immediately

Education

Nanyang Technological University

2026.08 – 2027.11 (Expected)

- MSc in Artificial Intelligence

University of Toronto

2021.09 – 2026.06

- BSc in Computer Science

Work Experience

TikTok | Software Developer Intern

2026.05 – 2026.08 (Expected)

- Optimized hot path data packing operators with flamegraph, decreasing average latency by 4.2ms, CPU utilization by 5%.
- Conducted thorough query and user feature migration, designed a diff test across 5 microservices and 647 features in total.

Tencent | Software Developer Intern

2025.05 – 2025.08

- Optimized an alert root-cause analysis agent using LangGraph, achieving 87% diagnostics accuracy and saving 15+ engineering hours per week.
- Designed a human-in-the-loop feedback pipeline to capture failure cases and refine prompts/workflows, enhancing alert resolution accuracy and accelerating iteration.
- Built an asynchronous proxy layer with thread-safe LRU-K cache for a user profile computation system, cutting storage cluster network bandwidth consumption by 43% and reducing read amplification by 79% for upstream RPCs.
- Implemented a cache-aware SIMD pooling operator for feature aggregation, reducing TP99 latency from 264μs to 93μs.

Urban Data Centre | AI/NLP Analyst

2024.09 – 2025.03

- Built a RAG pipeline for entity disambiguation in homonymous references, reducing hallucinated answer generations.
- Developed Python ETL pipelines to transform complex knowledge graph ontologies into XML/JSONL corpora, enabling a robust knowledge source for the RAG-based question answering.
- Fine-tuned a LLaMA-3-8B model with LoRA for English-to-SPARQL generation, replacing a rule-based baseline and improving query precision using schema-aware prompting.

JD.com, Inc. | Software Developer Intern

2024.05 – 2024.08

- Contributed to backend middleware for JD's shopping navigation search services on a large-scale e-commerce platform.
- Engineered a multi-threaded pipeline with rate limiting and concurrency control for incremental inverted-index loading, enabling real-time search updates.
- Refactored service routing registration into a unified API, eliminating duplicated configuration and simplifying setup effort for new search services.
- Authored technical documentation for modern C++ practices, adopted as a cross-team reference for intern onboarding.

University of Toronto | Teaching Assistant

2023.01 – 2026.04

- Delivered grading, office hours, and one-on-one tutoring for undergraduate computer science and mathematics courses.
- Supported 300+ students and provided 100+ hours of tutoring across Introduction to Computer Science and Linear Algebra.

Publications

- Lu, J. (2024). Parallel implementation of Wiener's attack on RSA: Algorithm design and performance evaluation. *Applied and Computational Engineering*, 37 184–190. <https://doi.org/10.54254/2755-2721/37/20230503>

Technical Skills

- **Programming Languages:** C, C++, Go, Python, Rust, Java, JavaScript, Bash, SQL.
- **Frameworks/Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, PEFT, scikit-learn, LangChain, LangGraph.
- **Tools/Systems:** Linux, Git, Docker, Redis, Kafka, MySQL, PostgreSQL, AddressSanitizer, Valgrind, pprof.
- **Languages:** English (Advanced), Chinese (Native).